

Chapter 8

Singular Value Decomposition

PCA is no more than a matrix factorization, which gives you new basis vectors and **scores** or coordinates for each observation:

$$\mathbf{X}^T = \mathbf{V}\mathbf{S}^T$$

- $\mathbf{V}$ is the matrix of eigenvectors (principal components)
- $\mathbf{S}^T$ contains the coordinates (scores) for each observation as columns.
- Covariance PCA: The matrix $\mathbf{X}$ contains centered data
- Correlation PCA: The matrix $\mathbf{X}$ contains centered and standardized data.

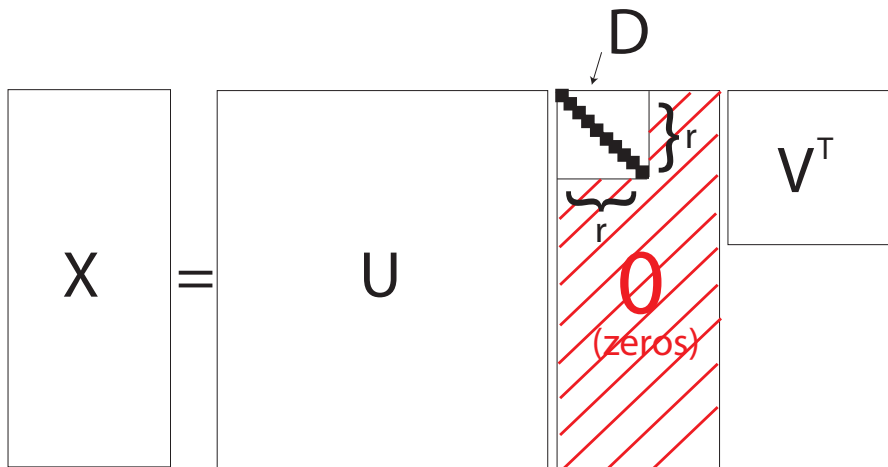
Above equation same as writing $\mathbf{X} = \mathbf{S}\mathbf{V}^T$.

Singular Value Decomposition

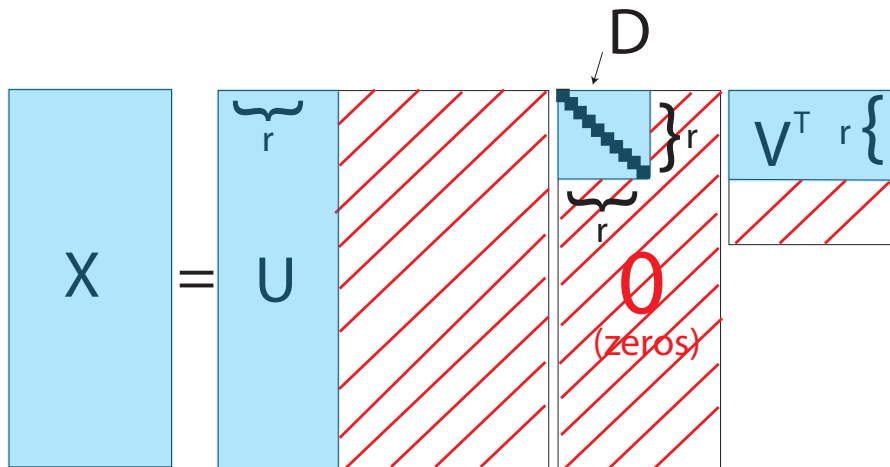
For any $n \times p$ matrix $\mathbf{X}$ with rank= r , there are orthogonal matrices $\mathbf{U}_{n \times n}$ and $\mathbf{V}_{p \times p}$ and a diagonal matrix $\mathbf{D}_{r \times r} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$ such that

$$\mathbf{X} = \mathbf{U} \underbrace{\begin{pmatrix} \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}}_{n \times p} \mathbf{V}^T \quad \text{with} \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r \geq 0$$

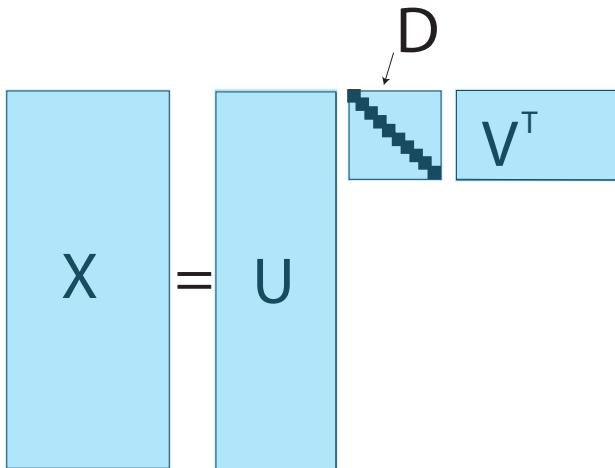
The SVD, illustrated



The SVD, illustrated



The “skinny” SVD, illustrated



Singular Value Decomposition

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T \quad \text{with} \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0$$

- The rows of $\mathbf{V}^T$ are orthonormal eigenvectors for $\mathbf{X}^T\mathbf{X}$
- The columns of $\mathbf{U}$ are orthonormal eigenvectors for $\mathbf{X}\mathbf{X}^T$
- $\sigma_i = \sqrt{\lambda_i}$
 - Diagonal matrix $\mathbf{D}$ contains the square roots of the eigenvalues
 - (the nonzero eigenvalues of $\mathbf{X}^T\mathbf{X}$ and $\mathbf{X}\mathbf{X}^T$ are the same)

Singular Value Decomposition

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T \quad \text{with} \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0$$

- The rows of $\mathbf{V}^T$ are called **right singular vectors**
- The columns of $\mathbf{U}$ are called **left singular vectors**
- σ_i 's are called **singular values**

If our data is centered... then $\mathbf{X}^T \mathbf{X}$ is the covariance matrix. So the rows of $\mathbf{V}^T$ are principal components! We already know that principal components analysis gives us

$$\mathbf{X} = \mathbf{S} \mathbf{V}^T$$

The SVD gives us

$$\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}^T$$

This means that the principal component scores are just scaled eigenvectors for $\mathbf{X} \mathbf{X}^T$:

$$\mathbf{S} = \mathbf{U} \mathbf{D}$$

Resolving a Matrix into Components

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T.$$

with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0$.

- We can think again of the σ_i 's as coordinates along some “basis matrices”.
- They tell us how much “information” from our data matrix $\mathbf{X}$ is directed along each basis “direction”.

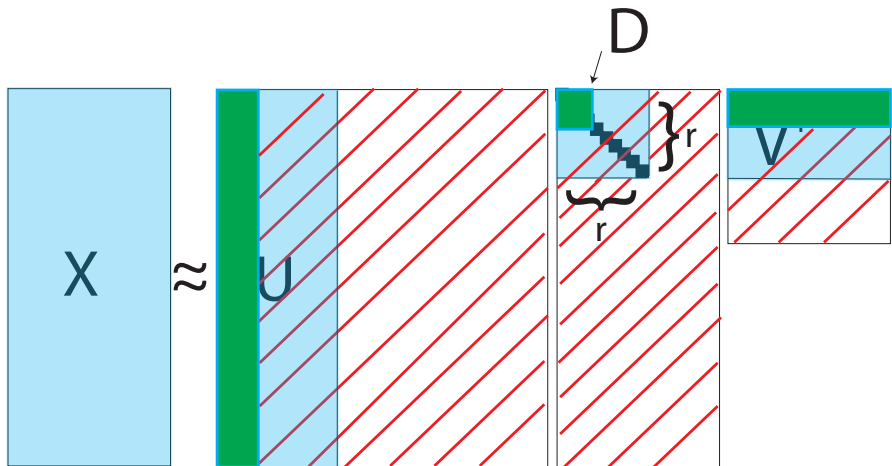
Resolving a Matrix into Components

$$\mathbf{X} = \mathbf{UDV}^T = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T.$$

with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0$.

- So the amount of “information” (or *signal*) is less in later components (as σ 's decrease).
- If we say “noise” is **uniformly distributed**, this means there is just as much noise along each direction.
- The “signal-to-noise” ratio is far less in later components. They are mostly noise.
- So let's chop them off!

The truncated SVD, illustrated



Resolving a Matrix into Fewer Components

This brings us to the **truncated SVD**, which only approximates the original data, but does so in the *best* way possible.

$$\mathbf{X} \approx \mathbf{U}_k \mathbf{D}_k \mathbf{V}_k^T = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_k \mathbf{u}_k \mathbf{v}_k^T.$$

- This gives us the *closest* rank k matrix to our original matrix x , which had rank $r > k$.
- Often times we can use a small percentage of the singular vectors to get a very good approximation to the data.
- This was an early form of image compression.